Year 1 Subject – Design Unit-1 Summative	 Image source: https://gict.co.in/wp-content/uploads/2024/07/Cover_Interaction-design.webp
	Designing for Interaction

Design Situation:

Young learners use many digital systems every day such as games, apps and learning platforms. These systems communicate with users through inputs, outputs and interactive responses.

In this unit, students will explore how simple **block-based coding** can be used to **design interactive digital solutions**. Through the **design cycle**, they will investigate user needs, develop ideas, create a simple interactive project and evaluate its effectiveness.

Students will create an **interactive project** using Pictoblox that demonstrates communication between a user and a digital system.

MYP 1 Unit-1 Elements

- **Specified Concepts:** Communication, Function, Systems, Innovation
- **Global Context:** Scientific and Technical Innovation (Products, Processes and Solutions)
- **Statement of Inquiry:** Understanding how systems and functions enable communication helps individuals design innovative interactive solutions that respond to user needs.
- **ATL Skills:** Thinking Skills, Self-management skills, Research skills, Communication Skills
- **IB Learner Profile:** Thinker

Command Terms

Command Term	Definition
Analyze	Break down in order to bring out the essential elements or structure. (To identify parts and relationships and to interpret information to reach conclusions.)
Create	To evolve from one's own thought or imagination, as a work or an invention.
Demonstrate	Make clear by reasoning or evidence, illustrating with examples or practical application.
Describe	Give a detailed account or picture of a situation, event, pattern or process.
Develop	To improve incrementally, elaborate or expand in detail. Evolve to a more advanced or effective state.
Evaluate	Make an appraisal by weighing up the strengths and limitations.
Explain	Give a detailed account including reasons or causes. (See also "Justify".)
Identify	Provide an answer from a number of possibilities. Recognize and state briefly a distinguishing fact or feature.
Justify	Give valid reasons or evidence to support an answer or conclusion. (See also "Explain".)
List	Give a sequence of brief answers with no explanation.
Outline	Give a brief account or summary.
Present	Offer for display, observation, examination or consideration.
State	Give a specific name, value or other brief answer without explanation or calculation.

THE GAUDIUM® SCHOOL
Sculpting happy minds

Middle Years Programme

A Description of Your Grasps Summative Assessment

Designing the product

G The goal is to design and create a simple interactive digital solution using Pictoblox that demonstrates how a digital system communicates with users through inputs, outputs and programmed responses.

R Your role will be a **young digital designer** responsible for creating an interactive project that engages users and demonstrates effective communication through coding.

A Your audience includes your classmates, teachers and young learners who will interact with and test your digital solution.

S You are in a situation that many digital systems require interaction between users and technology. Users provide information and expect meaningful responses. You have been challenged to design an interactive project that allows users to interact with a digital system using inputs, outputs and simple programmed logic. Your project should be easy to use, engaging and functional.

P The product is to design and create a guided interactive project using Pictoblox. Possible examples are:

- Guess the Number Game
- Simple Quiz
- Greeting Program
- Choice-based Interaction

S This stands for standards and criteria for success. Your work will be assessed using **all four MYP Design Criteria (A, B, C and D)**, focusing on research and analysis, idea development, creation and evaluation of your final coded solution.

Inquiry Questions

Factual: (Students need to know)

- What is a program?
- What are input and output?
- What are algorithms and flowcharts?

Conceptual: (Connect to specified concepts)

- How do systems and functions work in interactive programs?
- How does interaction improve communication?

Debatable: (To Encourage Discussion)

- Technology always improves communication

Contextual Lenses

Individual Lens

How do I interact with digital systems in my daily life?

Local Lens

How do people in my school or community use interactive applications?

Global Lens

How do interactive technologies impact communication worldwide?

Criteria A: Inquiring and Analysing

Young learners use digital systems such as games, apps and learning platforms every day. Interactive systems communicate with users through responses, feedback and user interaction. Your task is to create a simple interactive digital solution using PictoBlox that allows users to interact with the program.

Strand i: Explain and justify the need for a solution to a problem

Activity: Understanding the Problem

Research Skills – Information Literacy – Access information to be informed and inform others

Young learners use digital systems such as games, apps and learning platforms every day. These systems communicate with users through inputs, outputs and interactive responses.

a) Explain why interactive digital systems are important in everyday life.

____

____

____

____

b) Identify the users of your proposed interactive digital solution.

____

____

____

____

c) Explain how your solution could help, engage or communicate with users.

____

____

____

____

____

d) Justify why there is a need for your solution.

____

____

____

____

____

Strand (ii) State and prioritize the main points of research needed to develop a solution to the problem

Activity: Research Planning

Thinking Skills – Critical Thinking – Gather and organize relevant information to formulate an argument

To develop an effective interactive digital solution, identify the information you need to research.

a) State at least four things you need to research before creating your solution.

Examples:

- I am going to find out what features make an interactive program interesting for users.
- I am going to interview my classmates to learn what they expect from an interactive digital game.
- I am going to research which coding blocks can be used to make a program respond to user input.
- I am going to search for examples of simple interactive digital projects created by beginners.
- I am going to research how scores, points, or attempts can be tracked in a program.
- I am going to investigate what makes a digital solution easy to use and understand.

b) Prioritize the research points from most important to least important.

Priority	**Research Point**
Example:	I am going to interview my classmates to learn what they expect from an interactive digital game.
1	
2	
3	
4	

c) Explain briefly why your highest-priority research point is important.

____

____

____

____

Strand iii: Describe the main features of one existing product that inspire the solution to the problem.

Activity: Product Analysis

Research Skills – Media Literacy – Compare, contrast and draw connections among (multi)media resources

Investigate an existing interactive digital product.

Examples:

- Educational games
- Quiz applications
- Learning apps
- Interactive websites

For product analysis:

a) Identify the product.
b) Add an image, screenshot, or website link.
c) Describe the purpose/function of the product.
d) Identify its strengths.
e) Identify areas for improvement.
f) Identify one feature that inspires your own solution.
g) Explain how you could use that feature in your solution.

Students may use websites, apps, games, teacher-provided examples, observations, screenshots or videos as sources for their product analysis.

Criteria	Example	Product 1
Product Name	**Math Wizard**	
Image /Link	**https://plays.org/math-wizard/**	
Purpose / Function	**Helps users practise addition skills through a timed quiz game.**	
Strengths	• **Gives immediate feedback** • **Simple interface,** • **Easy to play** • **Includes a timer that makes the game exciting**	
Areas of Improvement	**It only focuses on addition and has limited game features.**	
Inspiring feature	**The game provides instant feedback after every answer.**	
How I Will Use This Feature in My Solution?	**I will make my program give immediate feedback so users know whether their answer or action is correct.**	

Strand iv: Present the main findings of the relevant research

Activity: Research Findings Summary

Communication Skills – Organize and depict information logically

Using your research and product analysis, summarize the most important information you learned.

Include:

- What users expect from an interactive digital game or program.
- Features that make a digital solution interesting and easy to use.
- Useful ideas you discovered from existing products.
- Coding techniques that may help your solution work successfully.
- Key information that will help you create your interactive digital solution.

Write a paragraph of approximately 50–100 words.

Criteria B: Developing Ideas

Before creating your interactive digital solution, you will generate ideas, develop success criteria, select the best design and create a plan for making your solution.

Strand i: Develop a list of success criteria for the solution

Activity: Success Criteria Development

Thinking Skills – Critical Thinking – Propose and evaluate a variety of solutions

Based on your research and design brief, develop a list of success criteria that your solution should meet. Your success criteria should help you judge whether your final solution is successful.

Examples:

- The game should be easy to understand and play.
- The user should be able to interact with the game.

Create at least **5 success criteria** for your solution.

NO.	Success criteria	Why is it important?
1		
2		
3		
4		
5		

Strand ii: Present feasible design ideas, which can be correctly interpreted by others

Activity: Generating Design Ideas

Thinking Skills – Creativity and Innovation – Use brainstorming and mind mapping to generate new ideas and inquiries

Develop and communicate different ideas for your interactive digital solution. Your ideas should clearly show:

- What the user will do
- How the program will respond
- The main interactive features

Present **two feasible design ideas**. For each idea include:

Design Idea 1

- Sketch / Screen Layout
- Description of interaction
- Inputs used
- Outputs provided

Design Idea 2

- Sketch / Screen Layout
- Description of interaction
- Inputs used
- Outputs provided

Design Idea 1	Design Idea 2
Description:	Description:

Strand iii: Present the chosen design

Activity: Choosing the Best Design

Thinking Skills – Critical Thinking – Evaluate evidence and arguments

Review your ideas and select the design that best meets your success criteria.

Task

a) Identify your chosen design.
b) Explain why you selected this design.
c) Explain how it meets your success criteria.
d) Explain why it is suitable for the intended users.

Criteria	Design Idea 1	Design Idea 2
Meets user needs		
Easy to create		
Interactive		
Meets Success Criteria		
Suitable for intended users		

Chosen Design Justification

____

____

____

____

____

Strand iv: Create a planning drawing/diagram which outlines the main details for making the chosen solution

Activity: Planning the Solution

Self-management Skills – Organization – Plan short and long-term assignments; meet deadlines

Create a detailed **flowchart** showing:

- Start
- Input
- Process
- Decision
- Output
- Stop

The flowchart should clearly show how the user interacts with the solution and how the program responds.

Criteria C: Creating the Solution

Before creating your final solution, you must develop a plan, use appropriate resources, build your interactive digital solution and record any changes made during the creation process.

Strand i: Outline a plan, which considers the use of resources and time, sufficient for peers to be able to follow to create the solution

Activity: Planning for Creation

Self-management Skills – Organization – Select and use technology effectively and productively

Before creating your solution, develop a step-by-step plan showing how you will build your interactive digital solution.

Your plan should include:

- Tasks to be completed
- Resources required
- Estimated time needed

Task: Complete the planning table below.

Step	Task to Complete	Resources Needed	Time Required
1			
2			
3			
4			
5			
6			
7			

Possible Resources:

- Computer/Laptop
- Pictoblox Software
- Flowchart
- Process Journal
- Teacher Guidance

Strand ii: Demonstrate excellent technical skills when making the solution
Activity: Developing the Solution
Thinking Skills – Transfer – Transfer current knowledge to learning of new technologies
Use Pictoblox to create your interactive digital solution.
Your solution should demonstrate appropriate use of:

- Sequencing
- Input and Output
- Variables
- Operators
- Conditional Statements (if / if-else)

Evidence of Technical Skills

- Paste screenshots or attach evidence showing:

Screenshot 1: Program Setup	
Screenshot 2: Input and Output Blocks	
Screenshot 3: Variables and Operators	
Screenshot 4: Conditional Statement	
Screenshot 5: Completed Program	

Strand iii: Follow the plan to create the solution, which functions as intended

Activity: Creating the Final Solution

Self-management Skills – Affective Skills – Demonstrate persistence and perseverance

Use your plan and flowchart to develop your solution.

Building Record: Describe the main steps you followed while creating your solution.

Step	Evidence	Explanation (What did you do?)
1		
2		
3		
4		
5		

Strand iv: List the changes made to the chosen design and plan when making the solution

Activity: Modifications During Creation

Self-management Skills – Reflection – Try new approaches to learning and evaluate their effectiveness

While creating your solution, you may have made changes to improve its functionality or usability.

Task: Record any changes you made.

Original Design/Plan	**Changes made**	**Reason for change**

Reflection on Changes

How did the changes improve your solution?

____

____

____

____

____

Criteria D: Evaluating

Before completing your project, you must test your solution, evaluate its success, identify possible improvements and reflect on its impact on users.

Strand i: Outline simple, relevant testing methods, which generate data, to measure the success of the solution

Activity: Testing the Solution

Thinking Skills – Critical Thinking – Test generalizations and conclusions

To find out if your solution works successfully, you must test it with users.

Task: Describe how you tested your solution.

____

____

____

____

Tester	Feedback
Friend	
Parent	
Classmate	

Test Conducted	Result
User entered input	
Program provided output	
Conditional statement worked correctly	
Program responded as expected	

Strand ii: Outline the success of the solution against the design specification

Activity: Evaluating Against Success Criteria

Self-management Skills – Reflection – Consider content (What did I learn? What don't I yet understand?)

Review the success criteria you created in Criterion B.

Success Criteria	Achieved? (Yes/No)	Evidence/Explanation
1		
2		
3		
4		
5		

Strand iii: Outline how the solution could be improved

Activity: Improvements

Thinking Skills – Creativity and Innovation – Design improvements to existing machines, media and technologies

If you had more time, what improvements would you make?

Improvement	Why would it help?

Strand iv: Outline the impact of the solution on the client / target audience

Activity: Impact on Users

Communication Skills – Give and receive meaningful feedback

Who is the target audience for your solution?

____

____

____

____

How did your solution help, engage or communicate with users?

____

____

____

____

____

____

____

What feedback did users give about your solution?

____

____

____

____

____

____

____

Reflection

What did you learn from completing this project?

____

____

____

____

____

____

What would you do differently next time?

____

____

____

____

____

____

Bibliography: (MLA9 format)

The bibliography lists all the sources you used during your research. MLA 9 is the referencing style commonly used in the IB Middle Years Programme.

Examples:

Book

- Resnick, Mitchel. *Lifelong Kindergarten: Cultivating Creativity through Projects, Passion, Peers, and Play.* MIT Press, 2017.

Website (PictoBlox)

- STEMpedia. *PictoBlox.* STEMpedia, https://thestempedia.com/product/pictoblox/. Accessed 27 June 2026.

Website (Scratch Ideas and Coding Concepts)

- Scratch Foundation. *Scratch.* https://scratch.mit.edu/. Accessed 27 June 2026.

Educational Website

- Code.org. *Learn Computer Science.* https://code.org/. Accessed 27 June 2026.

Online Tutorial

- STEMpedia. *PictoBlox Documentation.* https://ai.thestempedia.com/docs/pictoblox/. Accessed 27 June 2026.

Video

- STEMpedia. *Getting Started with PictoBlox.* YouTube, https://www.youtube.com/@STEMpedia. Accessed 27 June 2026.

Personal Interview

- Personal interview with your Design teacher about interactive digital systems, June 2026.

THE GAUDIUM SCHOOL
Sculpting happy minds

ib Middle Years Programme

Design Summative Assessment Rubric, Year1

Student Name: ______________________________

Criteria A: Inquiring and Analysing **Achievement Level: /8**

At the end of this unit 1, year 1 students should be able to:

i. Explain and justify the need for a solution to a problem.
ii. State and prioritize the main points of research needed to develop a solution to the problem
iii. Describe the main features of one existing product that inspires a solution to the problem.
iv. Present the main findings of the relevant research.

Achievement level	Achievement level descriptor	Task specific clarifications
0	The student **does not** reach a standard described by any of the descriptors below.	The student has provided insufficient evidence to demonstrate understanding of the problem, research, product analysis, or research findings.
1–2	The student: i. **states** the need for a solution to a problem ii. **states** the findings of research	• You have identified a simple need for an interactive digital solution. • You have identified users or basic information about the problem. • You have provided limited information about an existing digital product. • You have stated a few research findings related to your solution.
3–4	The student: i. **outlines** the need for a solution to a problem ii. **states some** points of research needed to **develop** a solution, **with some guidance** iii. **states** the main features of an existing product that inspires a solution to the problem iv. **outlines some** of the main findings of research	• You have explained the need for an interactive digital solution. • You have identified some research points that will help you develop your solution. • You have identified the purpose and some features of an existing digital product. • You have outlined some research findings that may help create your solution.
5–6	The student: i. **explains** the need for a solution to a problem ii. **states and prioritizes** the main points of research needed to **develop** a solution to the problem, with **minimal guidance** iii. **outlines** the main features of an existing product that inspires a solution to the problem iv. **outlines** the main findings of relevant research	• You have clearly explained why the solution is needed and how it will help users. • You have identified and prioritized relevant research activities such as interviewing users, researching coding blocks, exploring examples, or investigating interactive features. • You have described an existing product, including its purpose, strengths, areas for improvement, and an inspiring feature. • You have summarized relevant research

		findings that support the development of your solution.
7-8	The student: i. **explains and justifies** the need for a solution to a problem ii. **states and prioritizes** the main points of research needed to **develop** a solution to the problem, with **no guidance** iii. **describes** the main features of an existing product that inspires a solution to the problem iv. **presents** the main findings of relevant research	• You have fully explained and justified why the solution is needed and how it addresses the needs of users. • You have independently identified and prioritized relevant research activities required to create the solution. • You have thoroughly described the purpose, strengths, areas for improvement, and inspiring features of an existing product, and explained how the feature will influence your own solution. • You have presented clear, organized, and relevant research findings that directly support the development of your interactive digital solution.

Comments:

___

___

___

___

Criteria B: Developing Ideas **Achievement Level: /8**

At the end of this unit 1, year 1 students should be able to:

i. develop a list of success criteria for the solution
ii. present feasible design ideas, which can be correctly interpreted by others
iii. present the chosen design
iv. create a planning drawing/diagram which outlines the main details for making the chosen solution.

Achievement level	Achievement level descriptor	Task specific clarifications
0	The student **does not** reach a standard described by any of the descriptors below.	The student does not reach a standard described by any of the descriptors below.
1–2	The student: i. **states one** basic success criterion for a solution ii. **presents one** design idea, which can be interpreted by others iii. **creates** an **incomplete** planning drawing/diagram.	Your response shows that you have: • stated one simple success criterion for your interactive digital solution. • presented one basic idea with a simple sketch or description. • identified a chosen idea. • created an incomplete flowchart or planning diagram showing some parts of the solution
3–4	The student: i. **states a few** success criteria for the solution ii. **presents more than one** design idea, using an appropriate medium(s) or **labels key features**, which can be interpreted by others iii. **states** the key features of the chosen design iv. **creates** a planning drawing/diagram or **lists** requirements for the creation of the chosen solution.	Your response shows that you have: • stated a few success criteria based on user needs and project requirements. • presented more than one design idea using sketches, layouts or descriptions with labelled features. • identified key features of your chosen design. • created a simple flowchart or planning diagram showing how the solution will be created.
5–6	The student: i. **develops** a few success criteria for the solution ii. **presents** a few feasible design ideas, using an appropriate medium(s) and **labels key features**, which can be interpreted by others iii. **presents** the chosen design **stating** the key features iv. **creates** a planning drawing/diagram and **lists** the	Your response shows that you have: • developed clear and measurable success criteria linked to the design brief. • presented a few realistic and feasible design ideas with labelled features. • explained the key features of your chosen design and how it meets user needs. • created a clear flowchart or planning diagram showing inputs, processes, decisions and outputs needed to build the solution.

	main details for the creation of the chosen solution.	
7-8	The student: i. **develops a list of** success criteria for the solution ii. **presents** feasible design ideas, using an appropriate medium(s) and **outlines the key features,** which can be correctly interpreted by others iii. **presents** the chosen design **describing** the key features iv. **creates** a planning drawing/diagram, which **outlines** the main details for making the chosen solution.	Your response shows that you have: • developed a detailed list of clear and measurable success criteria based on research and user needs. • presented several feasible and well-developed design ideas with accurate labels and clear explanations. • justified your chosen design and described how its features meet the success criteria. • created a detailed, well-organized flowchart or planning diagram that clearly outlines the complete structure of the interactive digital solution, including inputs, processes, decisions and outputs.

Comments:

____

____

____

____

Criteria C: Creating the Solution **Achievement Level: /8**

At the end of this unit 1, year 1 students should be able to:

i. outline a plan, which considers the use of resources and time, sufficient for peers to be able to follow to create the solution
ii. demonstrate excellent technical skills when making the solution.
iii. follow the plan to create the solution, which functions as intended
iv. list the changes made to the chosen design and plan when making the solution

Achievement level	Achievement level descriptor	Task specific clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard described by any of the descriptors below.
1–2	The student: i. outlines a simple plan ii. demonstrates basic technical skills iii. creates a solution that functions partially iv. states a change made	Your work shows that: • You created a simple plan with a few basic steps. • You used some coding blocks with support. • Your program works only partially and may contain errors. • You identified one change made during creation.
3–4	The student: i. outlines a plan with some detail ii. demonstrates adequate technical skills iii. creates a solution that mostly functions as intended iv. outlines some changes made	Your work shows that: • You created a step-by-step plan including some resources and estimated time. • You used sequencing, input/output and other coding blocks correctly with some guidance. • Your interactive solution mostly works as planned. • You described some changes made during development and explained why they were needed.
5–6	The student: i. outlines a detailed and logical plan, sufficient for peers to follow ii. demonstrates excellent technical skills iii. follows the plan to create a solution that functions fully as intended iv. fully explains the changes made to the design and plan	Your work shows that: • You created a clear and logical plan that another student could follow. • You effectively used sequencing, input/output, variables, operators and conditional statements. • Your interactive solution functions as intended and meets most success criteria. • You clearly explained the changes made during development and how they improved the solution.

7-8	The student: i. outlines a detailed and logical plan, sufficient for peers to follow ii. demonstrates excellent technical skills iii. follows the plan to create a solution that functions fully as intended iv. fully explains the changes made to the design and plan	Your work shows that: • You created a detailed step-by-step plan including resources, time management and creation stages. • You confidently used sequencing, input/output, variables, operators and conditional statements to create an effective interactive solution. • Your final solution functions smoothly and fully meet the intended purpose and success criteria. • You clearly explained all modifications made during creation and justified how these changes improved the final solution.

Comments:

____

____

____

____

THE GAUDIUM SCHOOL
Sculpting happy minds

Middle Years Programme

Criteria D: Evaluating **Achievement Level: /8**

At the end of this unit 1, year 1 students should be able to:

i. outline simple, relevant testing methods, which generate data, to measure the success of the solution.
ii. outline the success of the solution against the design specification.
iii. outline how the solution could be improved.
iv. outline the impact of the solution on the client/target audience.

Achievement level	Achievement level descriptor	Task specific clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not demonstrate evidence of testing, evaluation, improvement, or impact.
1–2	The student: i. outlines one simple testing method ii. states whether the solution is successful or not iii. states one improvement iv. states one impact of the solution	Your response shows that you have: • outlined one simple testing method such as self-testing or asking one user to try the solution. • made a basic statement about whether the solution worked. • suggested one simple improvement. • stated one way the solution affected the user or audience.
3–4	The student: i. outlines a few simple, relevant testing methods ii. outlines the success of the solution against some design specifications iii. outlines a few improvements iv. outlines the impact of the solution on the client/target audience	Your response shows that you have: • used more than one testing method and recorded basic results or feedback. • compared your solution against some success criteria from Criterion B. • suggested a few realistic improvements. • described how the solution affected users using some feedback.
5–6	The student: i. outlines relevant testing methods that generate data ii. outlines the success of the solution against the design specification iii. outlines how the solution could be improved iv. outlines the impact of the solution on the client/target audience	Your response shows that you have: • used several relevant testing methods and collected clear results or user feedback. • evaluated your interactive solution against most of the success criteria. • suggested realistic improvements based on testing results. • clearly explained how the solution helped, engaged or communicated with the target audience.

7-8	The student: i. outlines detailed, relevant testing methods that generate data ii. outlines the success of the solution against the design specification iii. outlines how the solution could be improved iv. outlines the impact of the solution on the client/target audience	Your response shows that you have: • planned and carried out multiple relevant tests and collected meaningful user feedback and data. • clearly evaluated the success of your solution against all success criteria developed in Criterion B. • suggested well-reasoned and realistic improvements supported by evidence from testing. • clearly explained the impact of the solution on the target audience, supported by user feedback and testing results.

Comments:

____

____

____

____